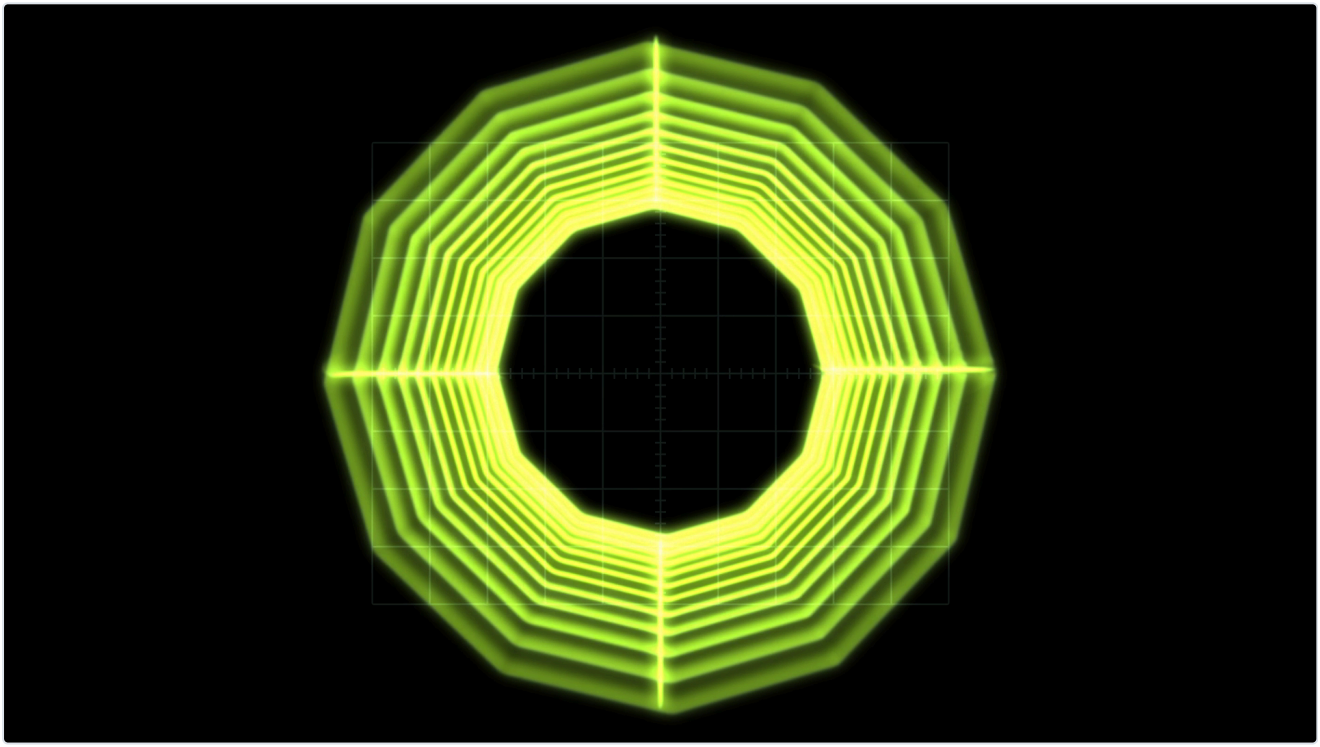
Vectrix user guide

Vectrix is **an oscillator, a pedalboard and a cathode ray tube in X/Y mode**, as two plugins for Resolume Arena and Avenue, and again as an OpenFX bundle for Resolve, Vegas, Nuke and Natron.

A function generator makes a two-channel signal. That signal goes through fourteen guitar and eurorack effects (a gate, a compressor, a wavefolder, a ring modulator, a phaser, a delay, a spring-ish reverb), and whatever comes out the far end drives the horizontal and vertical deflection of a scope. So what you're watching is where the beam went, rather than a picture of an oscilloscope.



> **Before you rely on this:** the main claim, that brightness follows dwell time, is measured rather than taken on trust. One sweep is rendered at ten speeds spanning a hundred to one, and the test fails if the total light deposited varies by more than half a per cent; a second test regresses the measured line density against `1/v`. Both test the same shader code that ships, and a control sweep fails if any parameter turns out to do nothing.
>
> **Both plugins have been loaded and run in Resolume Arena**, and since v0.1.5 both are also confirmed to load and register in Arena on Windows, which is worth saying out loud, because up to v0.1.4 they didn't. What's still unconfirmed is narrower: whether Arena's input is actually premultiplied, and how the Trace source's readback behaves for frame rate in a real host.
>
> **The OpenFX build of v0.1.6 to v0.1.11 crashed on every render**, in every OpenFX host, so if you tried one of those, update. v0.1.12 renders as a tool on Resolve's Fusion page; I haven't tried it on Resolve's other pages, or in Vegas, Nuke or Natron.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

It's inspired by Ms Mad Lemon's *Vector Synth Visualizer* series, which does the same thing properly, with 555 timers, LM358 op-amps and the deflection yoke of a CRT television.

Two plugins

- **Vectrix** (a source) generates its own signal over its own tube.
- **Vectrix Trace** (an effect) paints the clip below **on the tube face**, so it curves with the glass, sits behind the graticule and shares the halation. It can also take that clip *as* the signal, tracing its edges with the beam.

Both declare the same parameters, so a composition can be moved between them.

Start from a preset

Preset is the last group in the list, at the bottom of the panel. From v0.1.6 it's its own section rather than being tucked inside Output.

Each preset is a whole machine rather than a set of slider positions, so turning one knob on it does something you can predict. Start there and go outward.

There are a hundred and seventy-odd controls in the plugin. That's a lot, I know, but **the effect parameters *are* the plugin**, so cutting them wouldn't help. What v0.1.6 does instead is show you only the ones that can do something right now: about eighty. The Source you've chosen shows its own controls and not the other four sources', and each of the thirteen pedals shows only its on/off

switch until you switch it on, at which point its knobs appear. Turn Delay on and nine controls arrive with it.

Every pedal's switch is always visible, so there's never a state you can't get out of.

From v0.1.8 the controls that stand for a real quantity say so. Refresh Rate reads `64 Hz` rather than `0.8`, Delay Time reads `239 ms`, Gate Threshold reads `-40.2 dB`. Twenty-one controls do this: the ones that were always a frequency, a time or a level underneath, and only ever showed as a fraction because that's what the host asks a plugin for. Nothing about the sound or the picture changed; you can just see what you're setting now. Controls that are a proportion underneath, like Retrace Speed, still read as one.

Why brightness follows dwell time

Vectrix models the whole route from signal to glass, and every artefact comes from one fact.

The beam deposits light at a constant rate while it's on. Where it moves fast it spreads that light over a long distance and the trace is dim; where it turns around it lingers, and the trace blooms. **That's why a square drawn by an oscilloscope has bright corners and a square drawn by a computer doesn't.**

The renderer never computes `1/v`. It deposits a fixed quantum of energy per sample interval and spreads it over the distance the beam actually covered, and the `1/v` falls out of that, along with the right behaviour at both extremes and no clamps anywhere: a beam crossing the screen in one step is faint, and a beam standing still is bright but finite, because the energy is conserved either way.

Everything else follows the same rule, and it's worth knowing because it tells you what to expect:

- The **delay**'s repeats get rounder, because its feedback path is low-passed and a low-passed deflection signal has softer corners.
- The **gate** darkens the trace without moving it, because a grid voltage cuts beam current without touching the yoke.
- **Sample & Hold** draws lines between its steps, because the beam travels rather than teleporting.
- **Blank on Retrace** with two sawtooths gives a clean raster, because the beam is actually cut off during flyback. Nothing is skipping the flyback line.

Sources

Oscillator — sine, triangle, saw, ramp, pulse, noise and sample & hold per axis, with the X:Y ratio **locked to exact rationals** so a Lissajous figure stands still. Detune is a separate control, for when

you want the drift.

Shape — circle, square, diamond, polygon, star, rose, spirograph.

Wireframe — a spinning cube, a globe, a tunnel, and a scrolling three-dimensional landscape with hidden-line removal.

Audio file — WAV, AIFF, MP3, FLAC, Ogg and Opus. Left drives X, right drives Y. This is how oscilloscope music works, and Vectrix plays it. Mono files get a delayed-self plot rather than a flat line.

Trace (effect only) — finds the edges of the clip below and walks them with the beam. **Expect it to be excellent on titles, logos, graphics and line art, and a scribble on real footage.** That's down to the problem itself, and there isn't a setting that fixes it.

The pedalboard

Fixed order, because fourteen blocks is fourteen factorial orderings and "what order are my pedals in" is the first thing anyone gets wrong:

```
VCA → Gate → Compressor → Rectifier → Slew → Drive/Fold → Ring Mod → Bitcrush
    → Phaser → Flanger → Chorus → Delay → Reverb → Deflection Amp
```

Each block can process **both axes, one axis, or the mid/side pair**, and that routing control is where most of the pictures come from: an effect applied to one deflection axis and not the other stops being a filter and becomes a geometric transform.

Three worth going to first:

Phaser is the one that justifies doing any of this on the CPU. On X only it shifts each frequency component of the horizontal deflection by a different amount, so a sine figure *rotates* and a complex one *shears*. That isn't an affine transform, so no shader can produce it. At Mix 1.0 there are no notches at all and it becomes a pure rotator.

Compressor pumps the figure's *size*. It's the single best block for making a static figure feel alive, and it's what makes the audio-file source usable at all, since scope music has wildly varying level. Its limiter offers **Gain** (the figure shrinks) or **Rail** (the figure flattens against the edge of the screen, the way a deflection amplifier does when it runs out of headroom).

Slew limiter rounds every corner, and turns Sample & Hold from a constellation into a constellation with strings.

The tube

The phosphors come from a **table of measured data** rather than a tint.

P31 is the modern scope green, and at true persistence it shows no trail at all at 60 fps, because that's what P31 *is* (and why nobody photographs slow sweeps on one). **P7** is the two-layer cascade: a blue flash that pumps a yellow-green layer underneath, so the trail is a different colour from the strike.

Switching phosphor changes the brightness by up to three times, because the efficiencies are real. That isn't a bug to make up for with the master level.

Face Aspect, **Corner Radius** and **Deflection Gain** span both references without a second renderer: the defaults are a round lab scope, and 4:3 with a small corner radius and a little overscan gives the consumer television.

Cost: the face buffer is sized by the *spot*, not by the output, so 4K costs about two and a half times 1080p rather than four. What actually drives the cost is **Focus**: a sharper tube resolves more, and costs more for it.

In OpenFX hosts

Copy `Vectrix.ofx.bundle` to `/Library/OFX/Plugins` (macOS), `C:\Program Files\Common Files\OFX\Plugins` (Windows) or `/usr/OFX/Plugins` (Linux).

> **On Linux the plugin needs `libGL`, and nothing else unusual.** Vectrix's trace renderer runs on the GPU, so unlike its siblings this `.ofx` links OpenGL. GLEW is linked statically into the binary, so all the machine supplies is `libGL.so.1`, which is present on anything that can run Resolve and comes from the base repository on Rocky 8 / RHEL 8 (`mesa-libGL`).
>
> **v0.1.3 is the exception:** that build needs `libGLEW.so.2.0`, which on Rocky 8 means enabling PowerTools and installing `glew-devel`. See its release notes.
>
> The failure mode is silent either way: an OFX host doesn't report a plugin that failed to load, so a missing library looks like a broken download.

The signal chain is the same code (the engine is linked in, not reimplemented), but five things differ, and they're limitations rather than choices:

- **It renders on the CPU.** OpenFX hands a plugin no GL context, so the beam, the phosphor and the glass are a mirror of the GLSL rather than the GLSL. Offline-render speed, not interactive.
- **It replays.** Vectrix is stateful (phosphor, delay lines, a reverb tank), and OpenFX renders frames in whatever order it fancies. A seek, a jump or a parameter edit restarts the chain and warms it up over 120 frames. A linear render is exact; a reverb tail longer than that window isn't.

- **No tempo and no audio.** OpenFX has neither, so the tempo is pinned at 120 and the spectrum reads as silence. The LFOs still run, since they need only a frame duration, so the modulation slots are live.
- **The Trace source draws no beam**, because its edge pass needs a GPU. You get the tube, the graticule and the clip on the glass, with the gun cut off.
- **Resolve's Fusion page reports the frame rate on the effect but not on its clips.** The first OpenFX builds read a clip's rate, which Fusion leaves out, and failed every render there. Now Vectrix asks the output clip, the source clip and then the effect, so in Fusion the oscillator, the LFOs and the replay run at the timeline's own rate (checked at 24 and 25 fps). It assumes 24 fps, Resolve's default timeline rate, only where a host reports no rate at all.

If it looks wrong

The figure drifts instead of standing still. Detune isn't at zero. The X:Y ratio is locked to exact rationals precisely so it *can* stand still.

Everything got much darker when I changed phosphor. That's correct: the efficiencies are real, and P31 against P7 is up to a factor of three.

Trace on real footage is a scribble. Expected. Use it on titles, logos and line art.

No trail at all on P31. Also correct, and the reason P7 exists.

The effect does nothing. That's what a shader that wouldn't compile looks like from outside. Check the log: `~/Library/Logs/vectrix/` on macOS, `%LOCALAPPDATA%\vectrix\logs` on Windows, `~/.local/state/vectrix/logs` on Linux.

Resolume won't start at all on Windows, and deleting the plugin fixes it. A bug in v0.1.0 to v0.1.4 crashed the host while it was scanning the plugin folder (before any frame, and before the plugin could write a line to its own log), so there was nothing to look at. **Fixed in v0.1.5.** If you're on an earlier build, delete `Vectrix.dll` and `Vectrix Trace.dll` from `Documents\Resolume Arena\Extra Effects`, start Resolume to confirm it's happy, then install v0.1.5 or later. macOS was never affected.

In Resolve or another OpenFX host, Vectrix crashes or never renders a frame. A bug in v0.1.6 to v0.1.11 made the OpenFX build of both plugins segfault on every render, in every host. **Fixed in v0.1.12**: replace `Vectrix.ofx.bundle` with the one from the latest `-ofx-` download and restart the host. The Resolume build was never affected.
